

Module 3.3

Switching & Routing

Classification of Networks – based on scale

	Interprocessor distance	Processors located in same	Example
longer-range Networks	1 m	Square meter	Personal area network
	10 m	Room	Local area network
	100 m	Building	
	1 km	Campus	
	10 km	City	Metropolitan area network
	100 km	Country	Wide area network
	1000 km	Continent	
	10,000 km	Planet	The Internet (Internetwork)

Figure 1-6. Classification of interconnected processors by scale.

Personal Area Networks

- **Bluetooth-** A wireless network that connects a computer with its peripherals
- **Example1**

Bluetooth N/W's using master-slave paradigm

- PC (master)
- mouse, keyboard, etc. (slaves)
- The master tells the slaves what addresses to use, when they can broadcast, how long they can transmit, what frequencies they can use, and so on.

Personal Area Networks

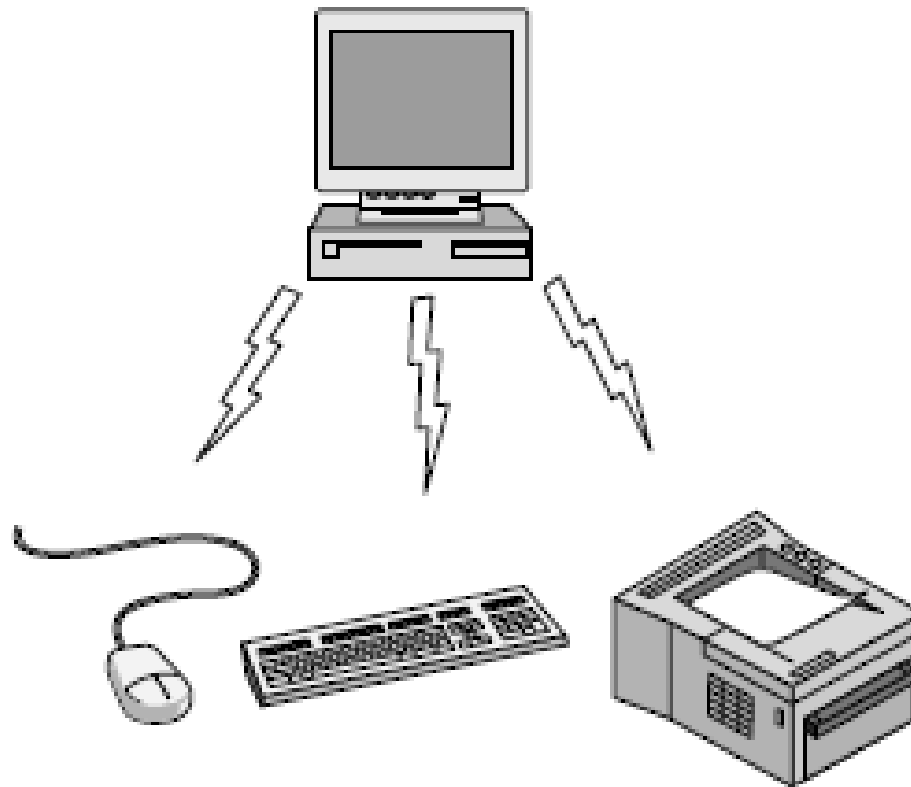
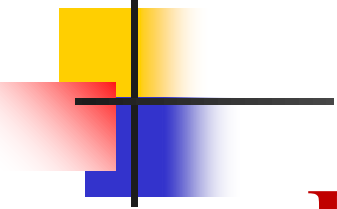


Figure 1-7. Bluetooth PAN configuration.



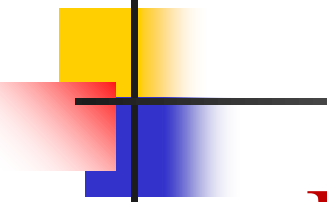
Personal Area Networks

- **Example2**

Bluetooth N/W to connect **headset** to a **mobile phone**

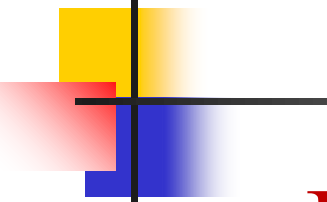
- **Example3**

A **PDA** controlling the user's **hearing aid** or **pacemaker**



Local Area Network(LAN)

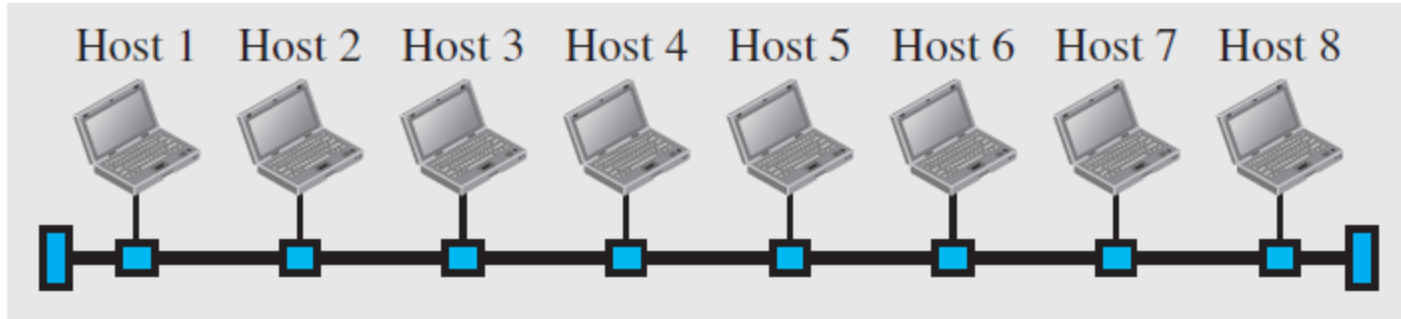
- Privately owned
- Links devices in a single home, office, building, or campus
- Allow resource sharing : i) **hardware** (e.g., a printer), ii) **software** (e.g., an application program), or iii) **data** to be shared between personal computers or workstations
- Distinguished based on **transmission media** and **topology**
- **Enterprise networks:** LANs used by companies



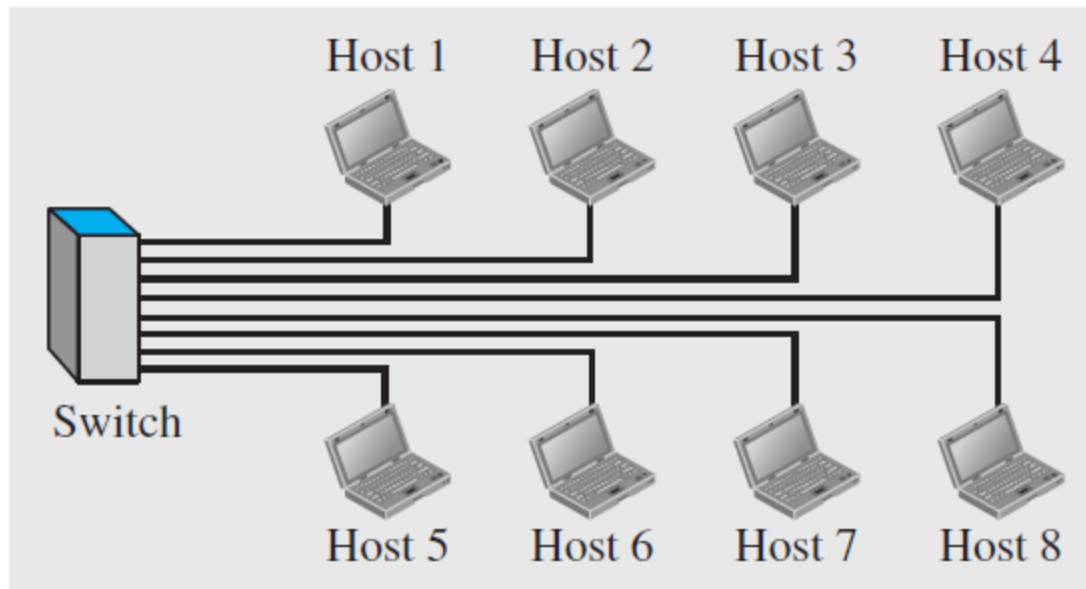
Local Area Network(LAN)

- Each **host** in a LAN has an **identifier** (an **address**), that uniquely defines the host in the LAN
- A **packet** sent by a host to another host carries both the **source host's address** and the **destination host's address**
- LAN with a **common cable** (a thing of the past)
- LAN with a **switch** (in use today)

Local Area Network(LAN)



a. LAN with a common cable (past)



b. LAN with a switch (today)

Legend



A host (of any type)



A switch



A cable tap



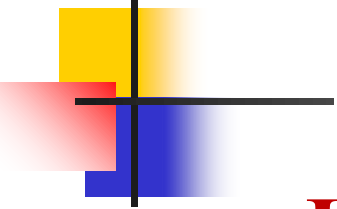
A cable end



The common cable



A connection



Local Area Network(LAN)-Wired LAN

- **Transmission media** - copper wires , optical fiber
- **Speeds** of 100 Mbps to upto 10 Gbps
- Low **delay** (μ s or ns) & very few errors
- **IEEE 802.3** (popularly called **Ethernet**) is by far the most common type of wired LAN

Local Area Network(LAN)-Wireless LAN

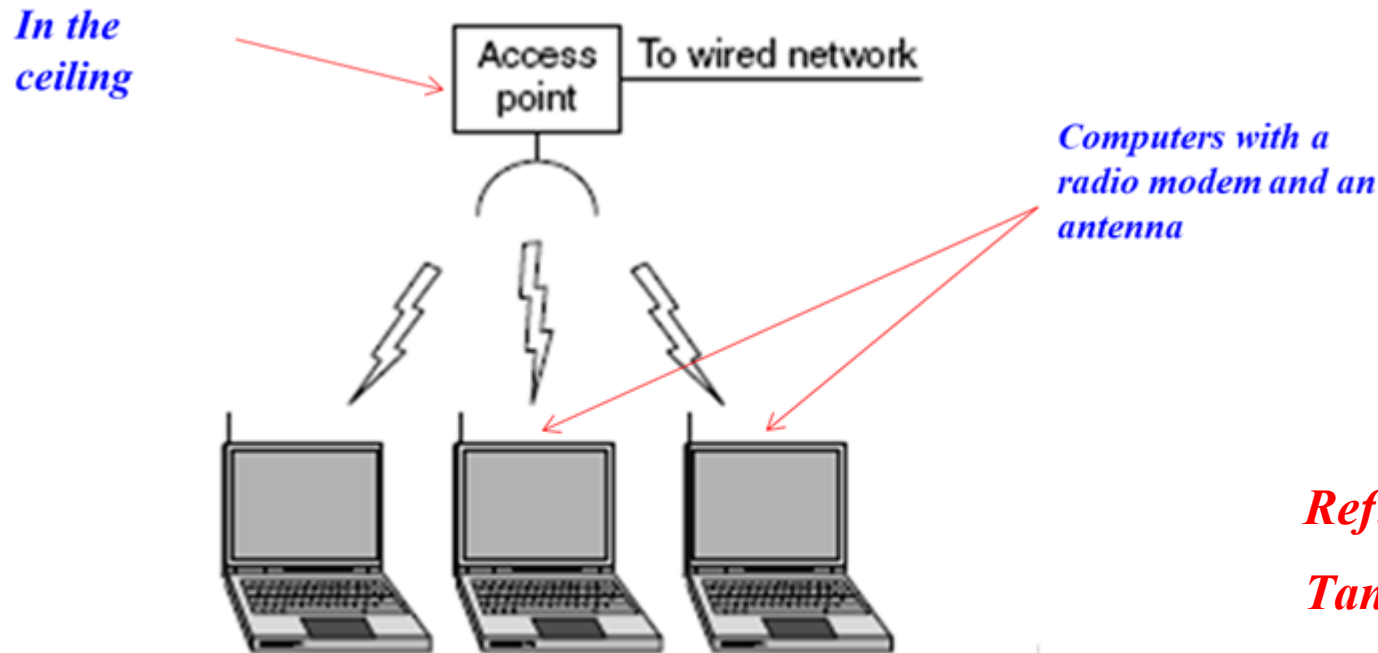
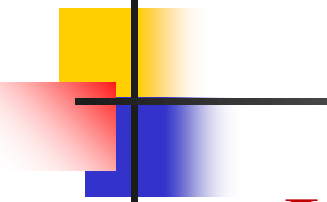


Figure 1-8. Wireless LANs. (a) 802.11.

Ref:
Tanenbaum

- *IEEE 802.11* (Popularly known as **WiFi**) - standard for wireless LANs.
- *Speed: 11 to hundreds of Mbps*

Forouzan, 5th



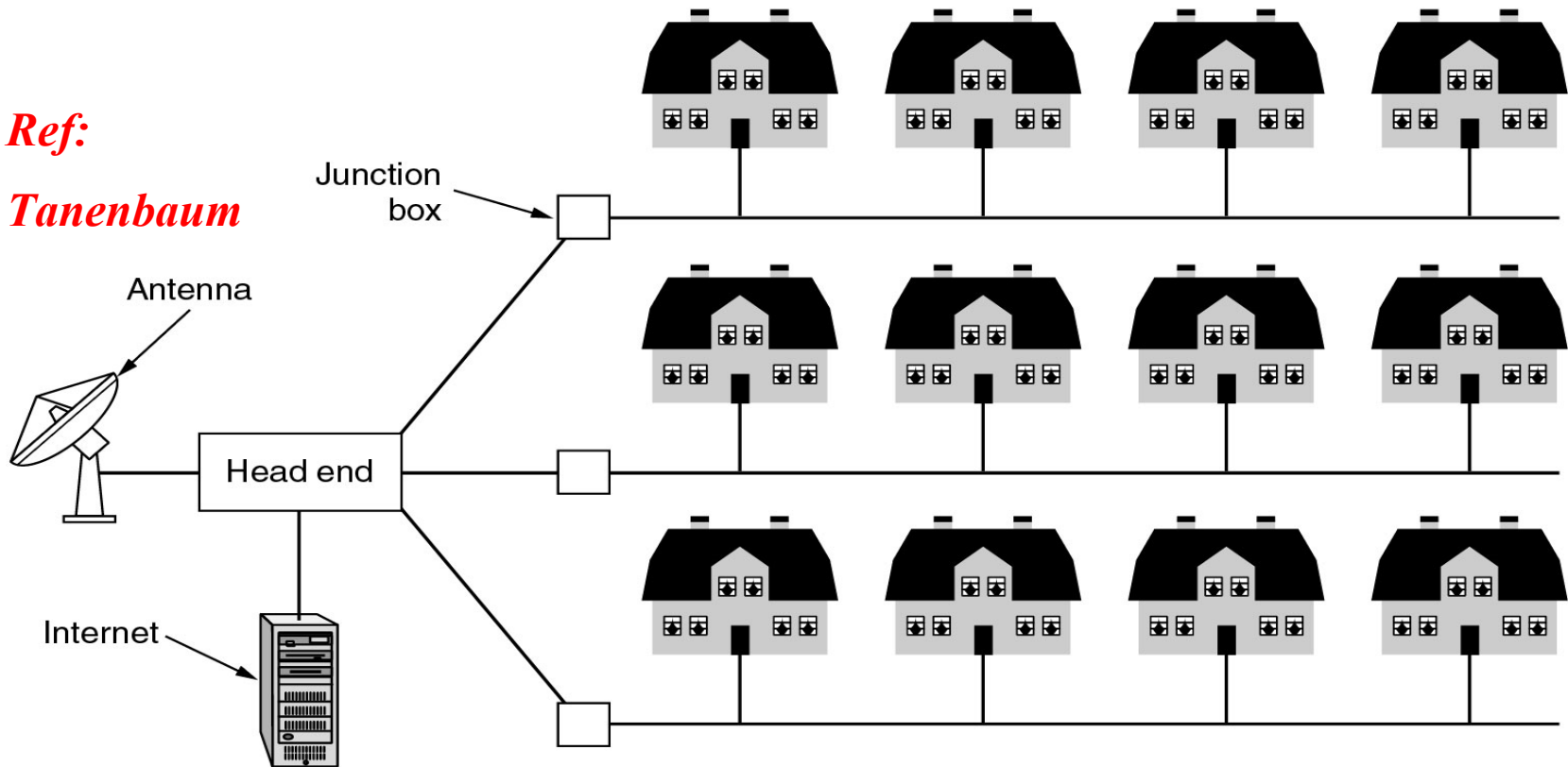
Local Area Network(LAN)-Virtual LAN

- Divide one large **physical LAN** into two **smaller logical LANs**
- **Why???**
 - the layout of the network equipment does not match the organization's structure

Metropolitan Area Networks

Ref:

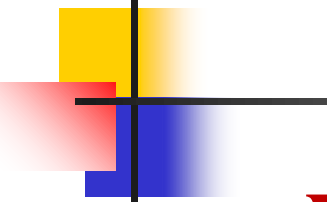
Tanenbaum



Examples

- *A MAN based on cable TV.*
- *For high-speed wireless Internet access, IEEE 802.16 (popularly known as **WiMAX**, Worldwide Interoperability for Microwave Access)*

Basis	LAN	MAN	WAN
Full-Form	LAN stands for local area network.	MAN stands for metropolitan area network.	WAN stands for wide area network.
Geographic Span	Operates in small areas such as the same building or campus.	Operates in large areas such as a city.	Operates in larger areas such as country or continent.
Ownership	LAN's ownership is private.	MAN's ownership can be private or public.	While WAN also might not be owned by one organization.
Transmission Speed	The transmission speed of a LAN is high.	While the transmission speed of a MAN is average.	Whereas the transmission speed of a WAN is low.
Propagation delay	The propagation delay is short in a LAN.	There is a moderate propagation delay in a MAN.	Whereas, there is a long propagation delay in a WAN.
Congestion	There is less congestion in LAN.	While there is more congestion in MAN.	Whereas there is more congestion than MAN in WAN.
Design & Maintenance	LAN's design and maintenance are easy.	While MAN's design and maintenance are difficult than LAN.	Whereas WAN's design and maintenance are also difficult than LAN as well MAN.
Fault tolerance	There is more fault tolerance in LAN.	While there is less fault tolerance.	In WAN, there is also less fault tolerance.



Wide Area Network

- **Point-to-point WAN**

- A network that **connects two communicating devices** through a transmission media (**cable or air**)

- **Switched WAN**

- A network with **more than two ends**
- A switched WAN is a combination of **several point-to-point WANs** that are connected by **switches**.
- It is used in the **backbone of global communication today**

Figure 1.11 *WANs: a point-to-point WAN & a switched WAN*

Figure 1.9 *A point-to-point WAN*

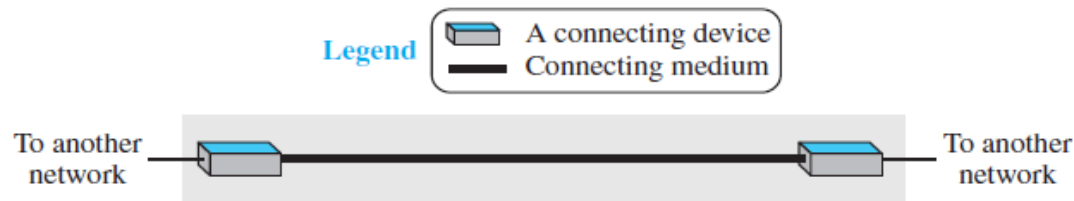
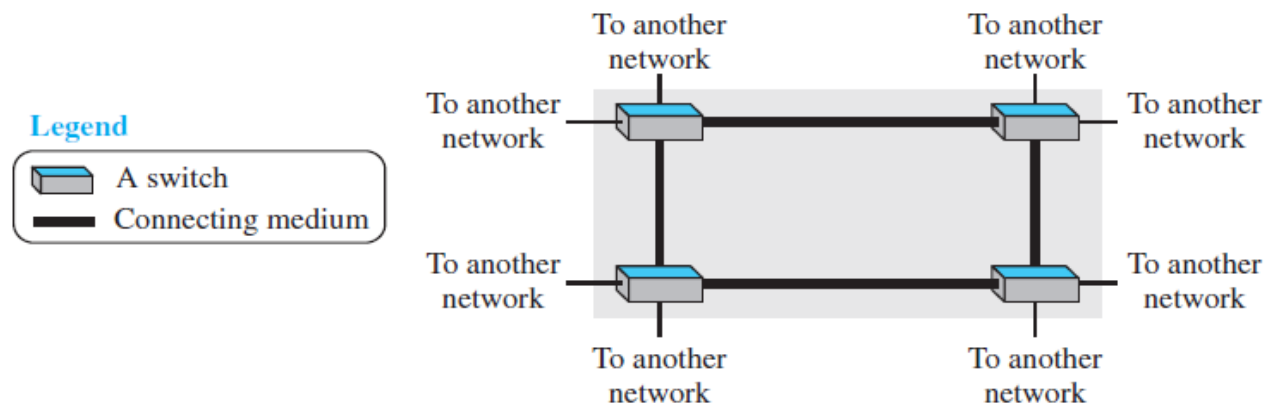
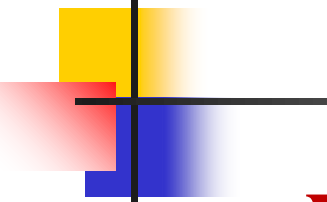


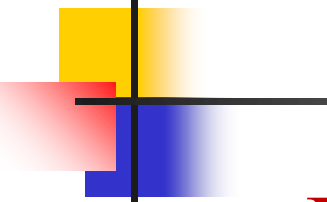
Figure 1.10 *A switched WAN*





Wide Area Network

- WAN looks like a large wired LAN but is it just that?
- **Differences:**
 1. A **LAN is normally limited in size**, spanning an office, a building, or a campus; a WAN has a wider geographical span, spanning a town, a state, a country, or even the world
 2. A **LAN interconnects hosts**; a WAN interconnects connecting devices such as switches, routers, or modems
 3. A **LAN is normally privately owned** by the organization that uses it; a WAN is normally created and run by communication companies and leased by an organization that uses it.

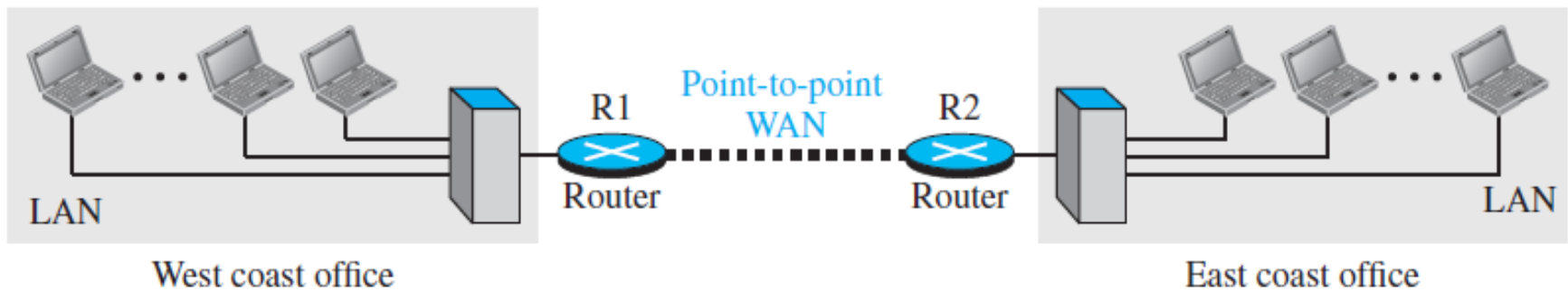


Internetworks or internet (small 'i')

- **internet:** Collection of (different, and frequently incompatible) interconnected networks
- As an example (*Fig 1.11*)
 - assume that an organization has two offices, one on the east coast and the other on the west coast.
 - Each office has a LAN that allows all employees in the office to communicate with each other
 - To make the communication between employees at different offices possible, the management leases a point-to-point dedicated WAN from a service provider, such as a telephone company, and connects the two LANs
 - Now the company has an internetwork, or a private internet (with lowercase *i*)

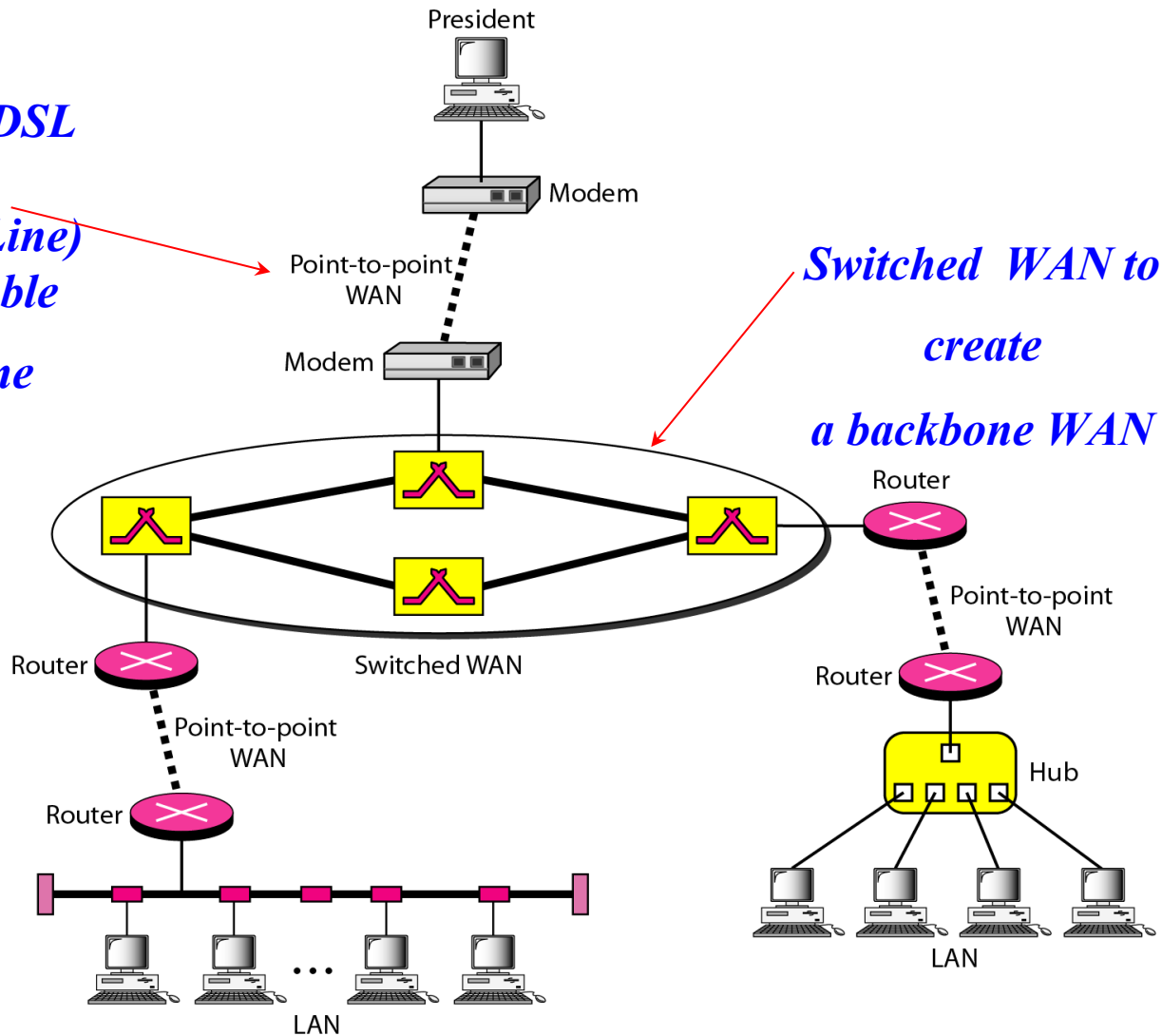
Internetworks or internet (small 'i')

Figure 1.11 *An internetwork made of two LANs and one point-to-point WAN*



A heterogeneous network made of four WANs and two LANs

*high-speed DSL
(Digital
Subscriber Line)
line or a cable
modem line*



Switching

Why Switching?

- one-to-one communication between each pair of devices are impractical and wasteful when applied to very large networks
- **Switches (interlinked nodes)** are devices capable of creating **temporary connections** between two or more devices linked to the switch

Figure 8.1 Switched network

A, B, C, D,... □ End systems

I, II, III, IV, and V □ Switches

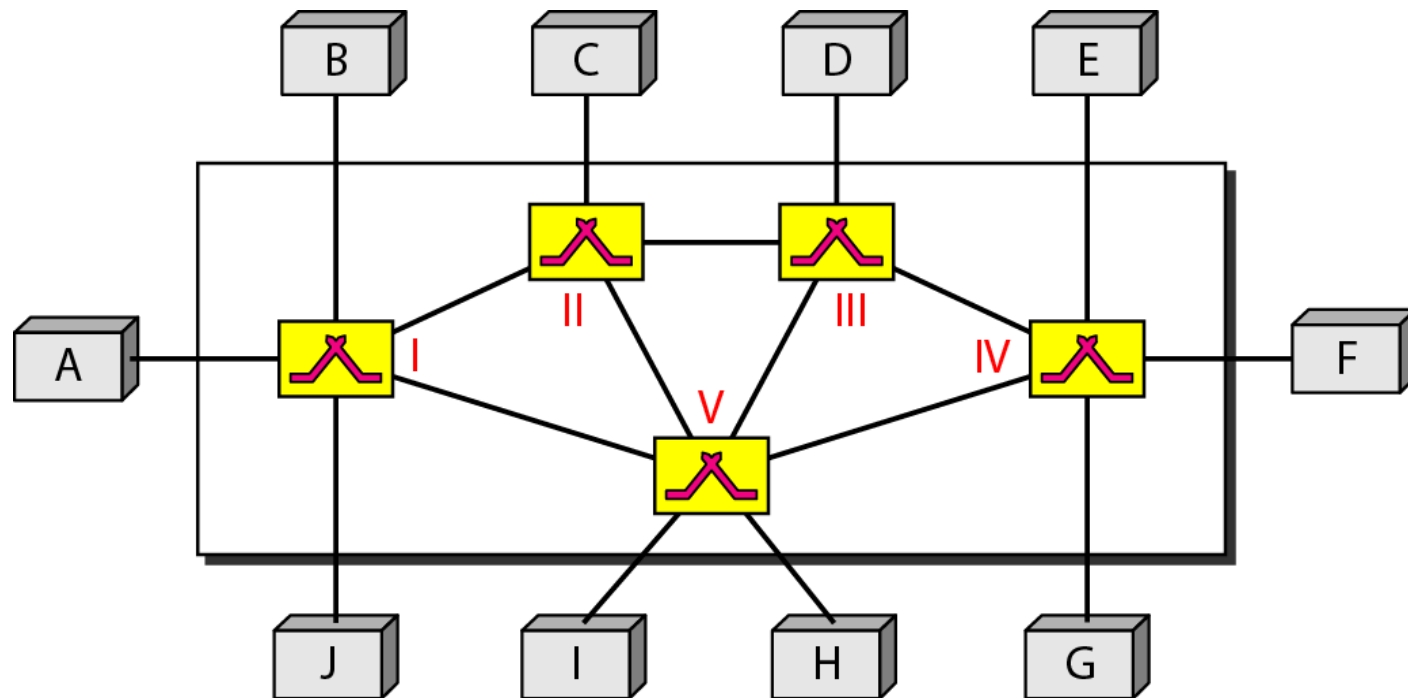
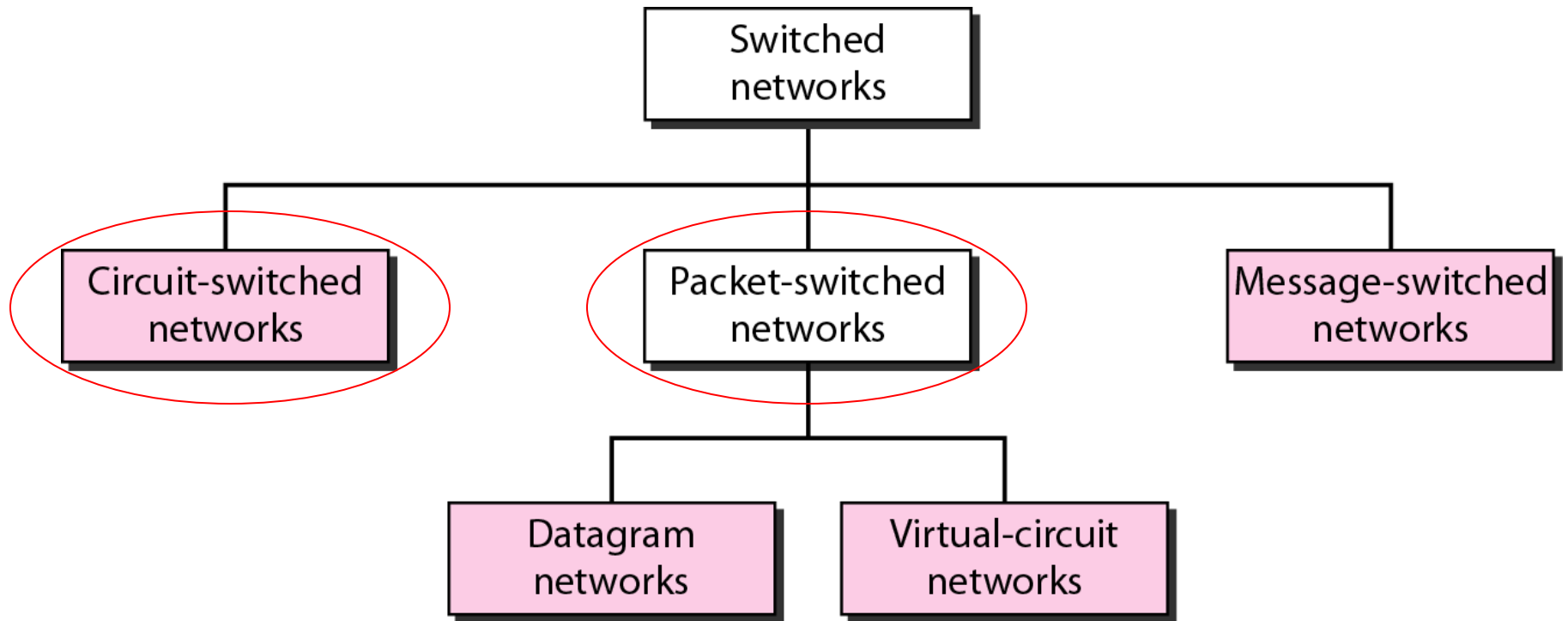


Figure 8.2 Taxonomy of switched networks



Taxonomy of switched networks

The order -

1. circuit-switched networks
2. datagram networks
3. virtual-circuit networks (has some common characteristics of 1 & 2)

8-1 CIRCUIT-SWITCHED NETWORKS

- *A circuit-switched network consists of a set of switches connected by physical links.*
- *A connection between two stations is a dedicated path made of one or more links.*
- *Each link is normally divided into n channels by using FDM (Frequency Division Multiplexing) or TDM (Time Division Multiplexing).*
- *Each connection uses only one dedicated channel on each link.*

Note

A circuit-switched network is made of a set of switches connected by physical links, in which each link is divided into n channels.

Figure 8.3 A *trivial* circuit-switched network

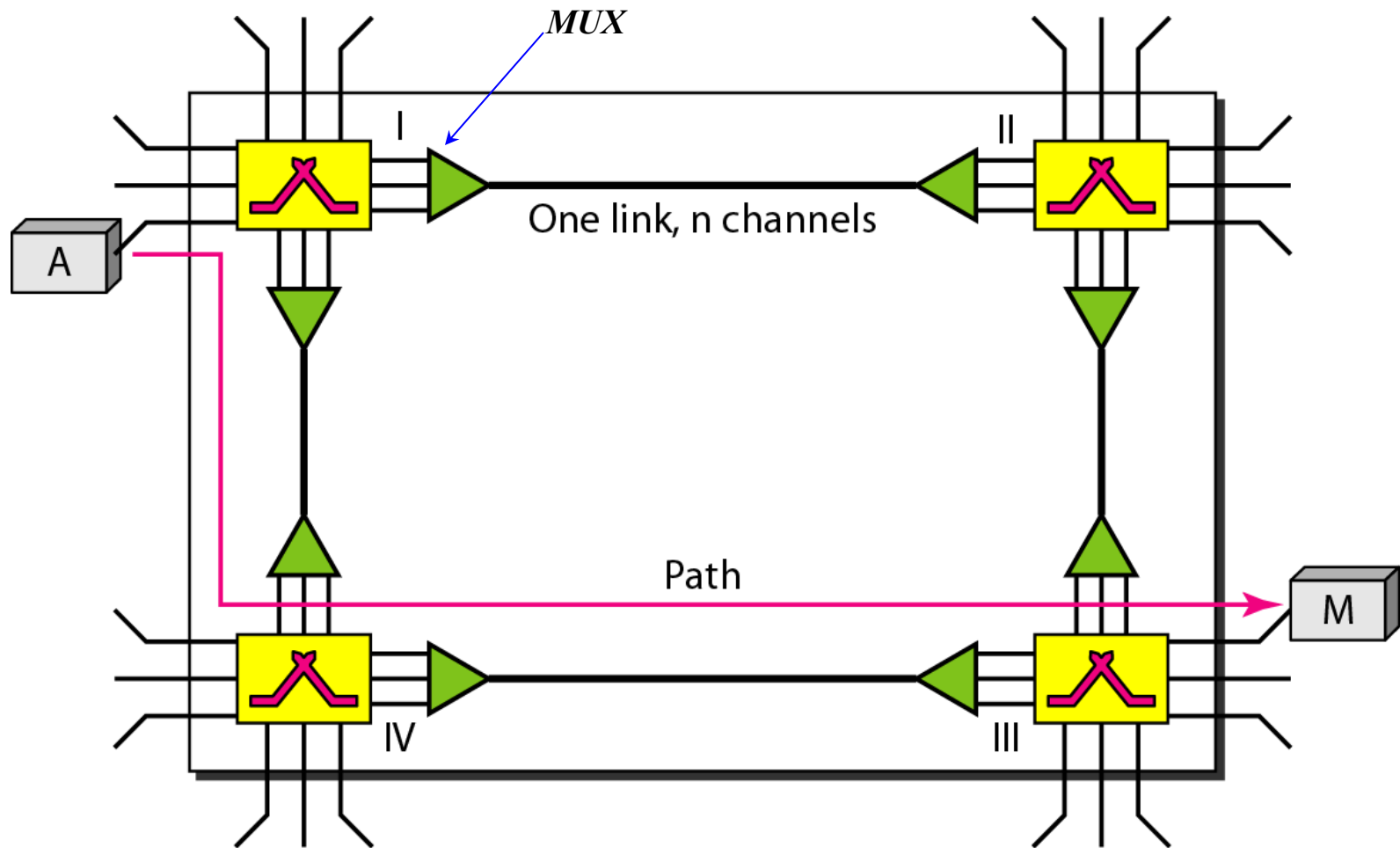


Figure 8.3 *A **trivial** circuit-switched network*

A needs to communicate with M

- ❑ A needs to request a connection to M that must be accepted by all switches as well as by M itself - **setup phase**.
- ❑ **dedicated path** of circuits (channels) is established by reserving a circuit (channel) on each link.
- ❑ **Data transfer** takes place.
- ❑ After all data have been transferred, the circuits are **torn down**.

Figure 8.3 *A **trivial** circuit-switched network*

- Circuit switching takes place at the **physical layer**
- Stations must make a **reservation** for the dedicated resources **until the teardown phase**
- **Data** transferred between the two stations are **not packetized**
- There is **no addressing** involved during data transfer (**end-to-end addressing used during the setup phase**)

Note

In circuit switching, the resources need to be reserved during the setup phase; the resources remain dedicated for the entire duration of data transfer until the teardown phase.

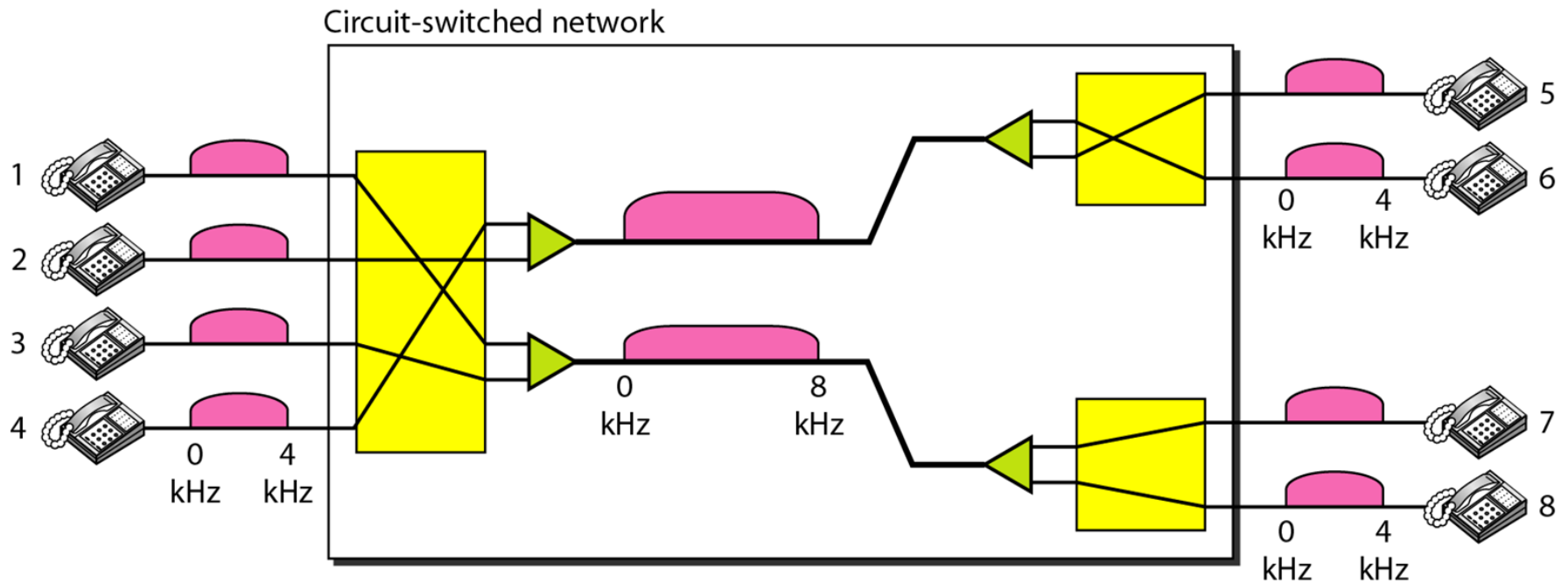


Example 8.1

As a trivial example, let us use a circuit-switched network to connect eight telephones in a small area.

- *Communication is through 4-kHz voice channels.*
- *We assume that each link uses FDM to connect a maximum of two voice channels.*
- *The bandwidth of each link is then 8 kHz.*
- *Telephone 1 is connected to telephone 7; 2 to 5; 3 to 8; and 4 to 6.*
- *The switch controls the connections.*

Figure 8.4 *Circuit-switched network used in Example 8.1*



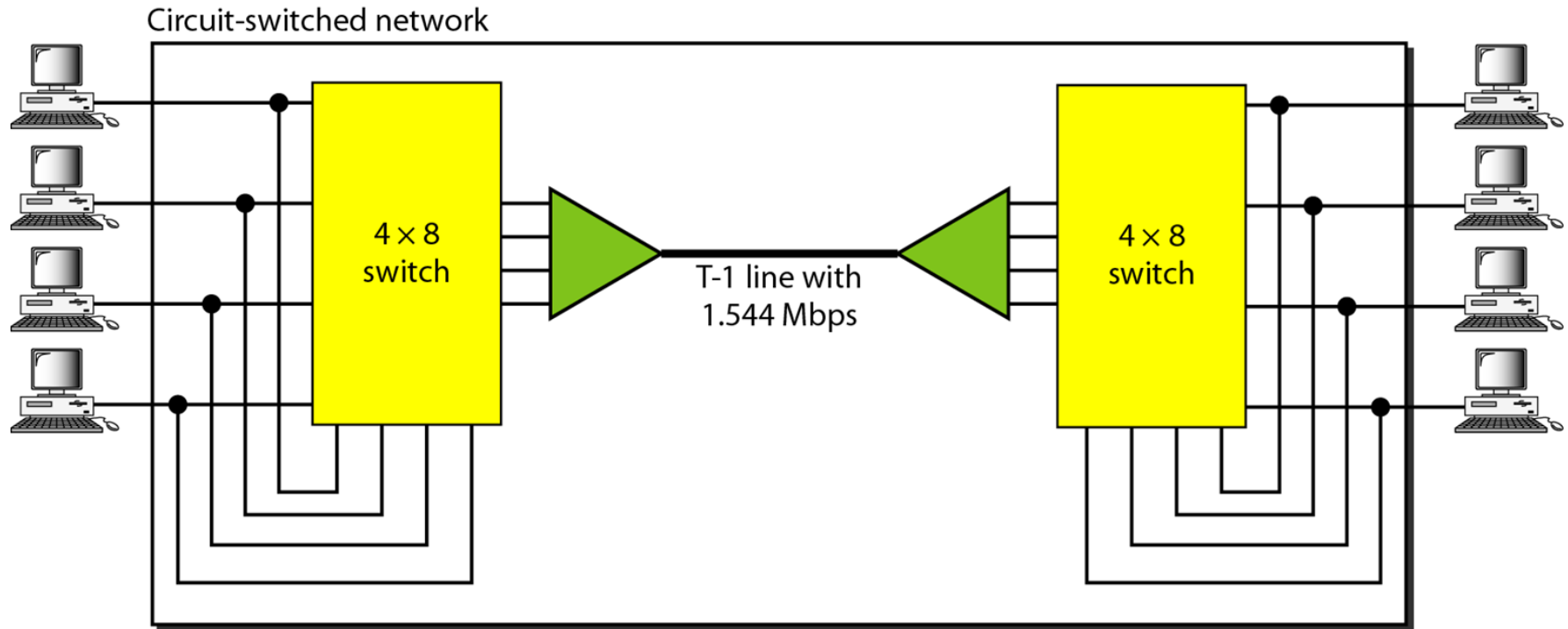


Example 8.2

*As another example, consider a **circuit-switched network** that connects computers in two remote offices of a private company.*

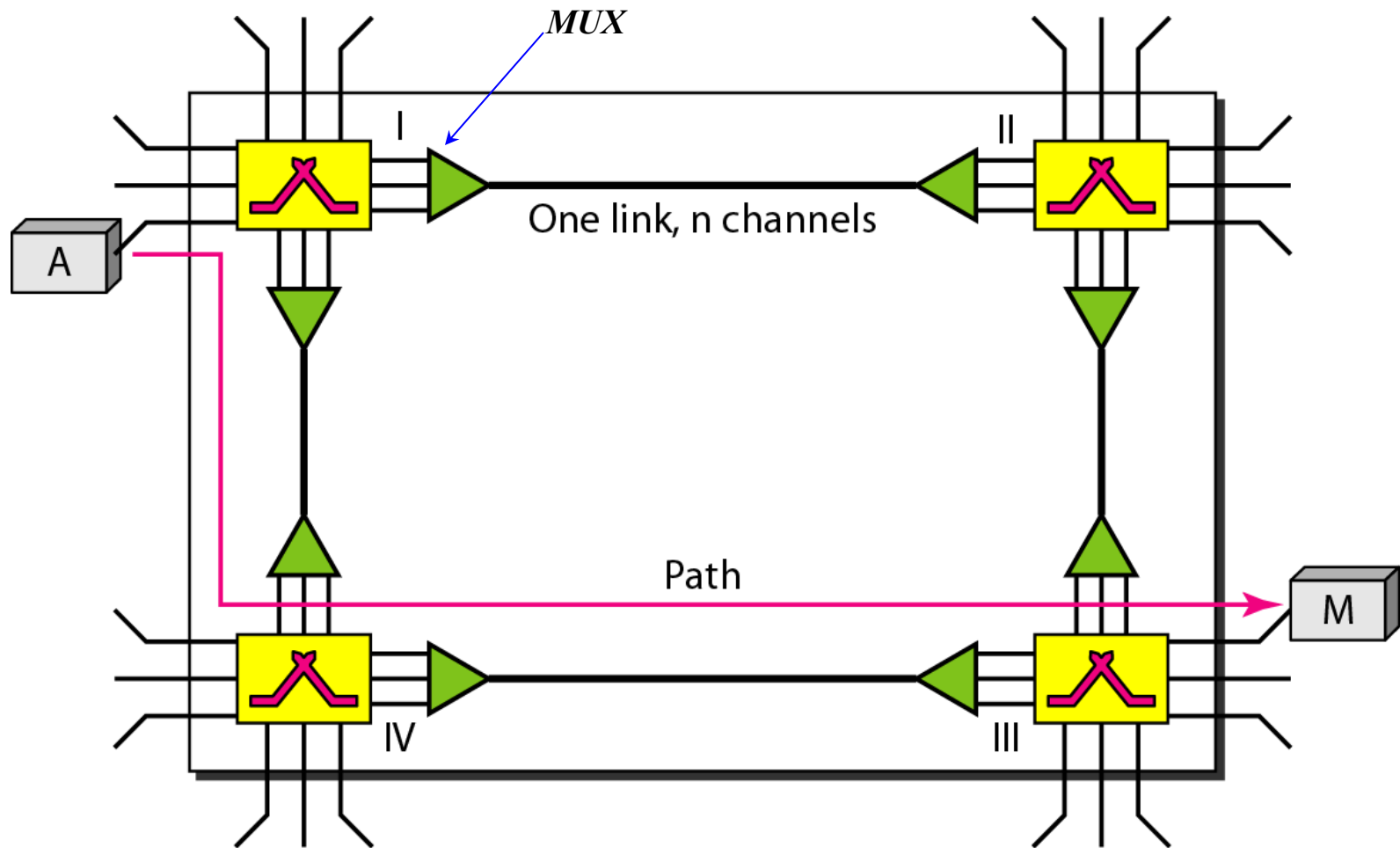
- *The offices are connected using a **T-1 line** leased from a communication service provider.*
- *There are **two 4×8 (4 inputs and 8 outputs) switches** in this network.*
- *For each switch, **four output ports are folded into the input ports** to allow communication between computers in the same office.*
- *Four other output ports allow communication between the two offices. **Figure 8.5** shows the situation.*

Figure 8.5 *Circuit-switched network used in Example 8.2*



T1: 24 simultaneous phone calls, that provides a dedicated, reliable data rate of 1.544 Megabits per second (Mbps), symmetrical for uploads and downloads, commonly used by businesses for consistent internet and voice (VoIP).

Figure 8.3 A *trivial* circuit-switched network



Three Phases

1. Connection setup:

- ❑ creating dedicated channels between the switches

In the previous example: (**Fig 8.3**)

- ✓ **A** sends a setup request (including the address of system **M**) to **switch I**
- ✓ **switch I** sends the request to **switch IV**
- ✓ **switch IV** sends a request to **switch III**
- ✓ **switch III** informs system **M**
- ❑ an **acknowledgment** from **system M** needs to be sent to **system A**
- ❑ end-to-end addressing is required for creating a connection between the two end systems (**Ex. telephone numbers in an FDM network**)

Three Phases

2. data transfer phase

3. connection teardown

- ✓ When one of the parties needs to **disconnect**, a signal is sent to each switch to release the resource.

Efficiency (LOW)

- Resources are allocated during the **entire duration** of the connection
- ✓ a computer can be connected to another computer even if there is **no activity for a long time**

Delay

- Delay is minimal
- ✓ resources are allocated(dedicated) for the duration of the connection □ no delay at each switch
- ✓ Example for delay in next slide!

Note

Switching at the physical layer in the traditional telephone network uses the circuit-switching approach.

8-2 DATAGRAM NETWORKS

- *In data communications, we need to send **messages** from one end system to another.*
- *If the message is going to pass through a packet-switched network, it needs to be **divided into packets of fixed or variable size**.*
- *The size of the packet is determined by*
 1. *the network and*
 2. *the governing protocol.*

DATAGRAM NETWORKS

- **No resource allocation** for a packet (**no** reserved **bandwidth** on the links & **no** **scheduled processing time**)
- Allocation is done on a first-come, **first-served basis**
- When a switch receives a packet the **packet must wait** if there are other packets being processed
- Packets are treated **independently** even if part of a multipacket transmission!

Note

*In a packet-switched network, there
is no resource reservation;
resources are allocated on demand.*

Figure 8.7 *A datagram network with four switches (routers)*

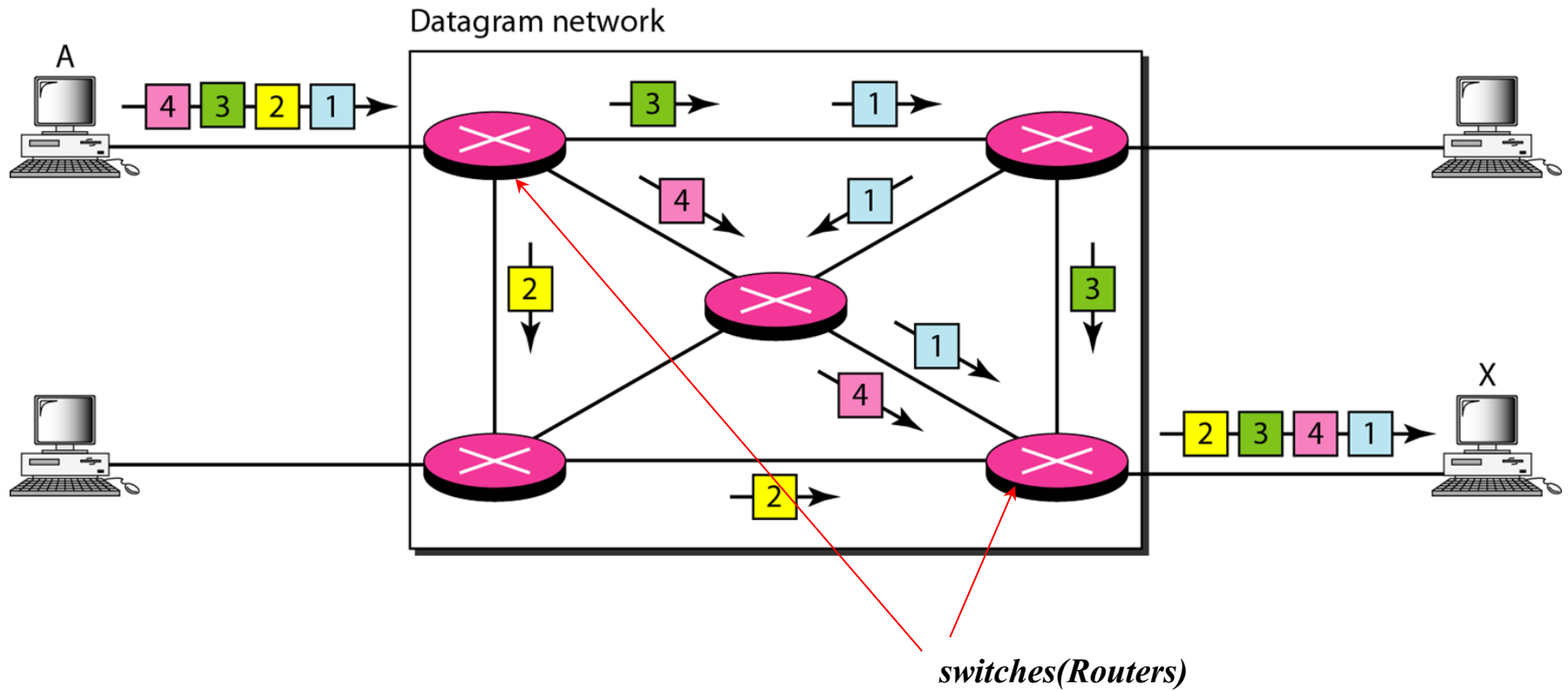


Figure 8.7 *A datagram network with four switches (routers)*

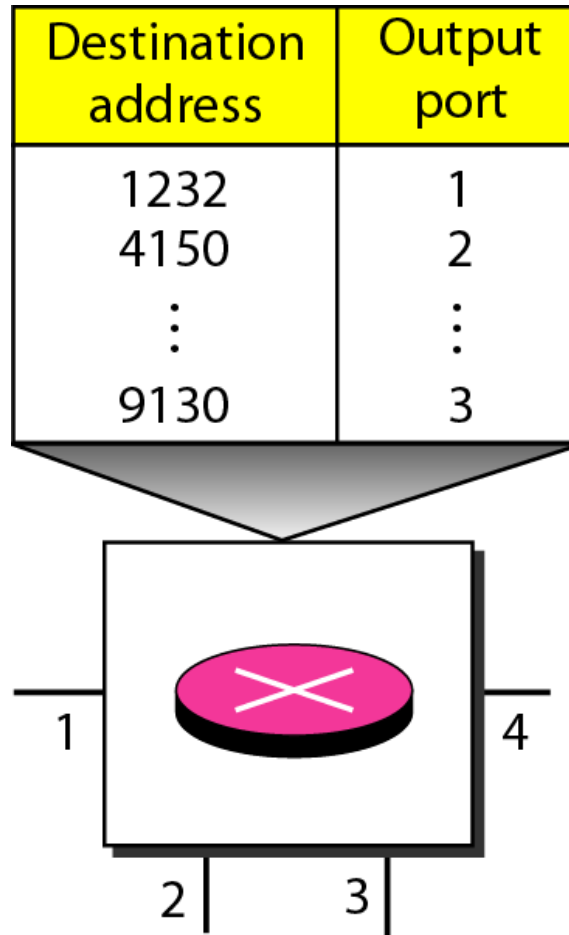
- All **four packets** (or datagrams) belong to the same message, but may **travel different paths** to reach their destination.
- Datagrams arrive at their destination **out of order**
- Packets may also be **lost** or **dropped**
- It's the responsibility of an **upper-layer protocol** to **reorder** the datagrams or **ask for lost datagrams**
- **It's a Connectionless network** (no set up or teardown phases!)

DATAGRAM NETWORKS

- Routing Table

- Each packet switch has a **routing table** which is based on the **destination address**.
- The routing tables are **dynamic** and are updated periodically.
- The **destination addresses** and the corresponding **forwarding output ports** are recorded in the tables.
- Every packet carries a **header** that **contains the destination address** of the packet.
- When the switch receives the packet, **destination address is examined**; the **routing table is consulted** & port through which the packet should be forwarded is found

Figure 8.8 *Routing table in a datagram network*





A switch in a datagram network uses a routing table that is based on the destination address.



The destination address in the header of a packet in a datagram network remains the same during the entire journey of the packet.

Efficiency (**Better**)

- **Resources allocated** only when packets are to be transferred.
- If **there is a delay** (minutes) b/w packets from a source, **resources can be reallocated** during these minutes for **other source packets**

Delay

Greater than virtual-circuit N/W

✓ each packet may experience a wait at a switch before it is forwarded

✓ **Delay** is **not uniform** for the packets of a message (packets travel different paths)



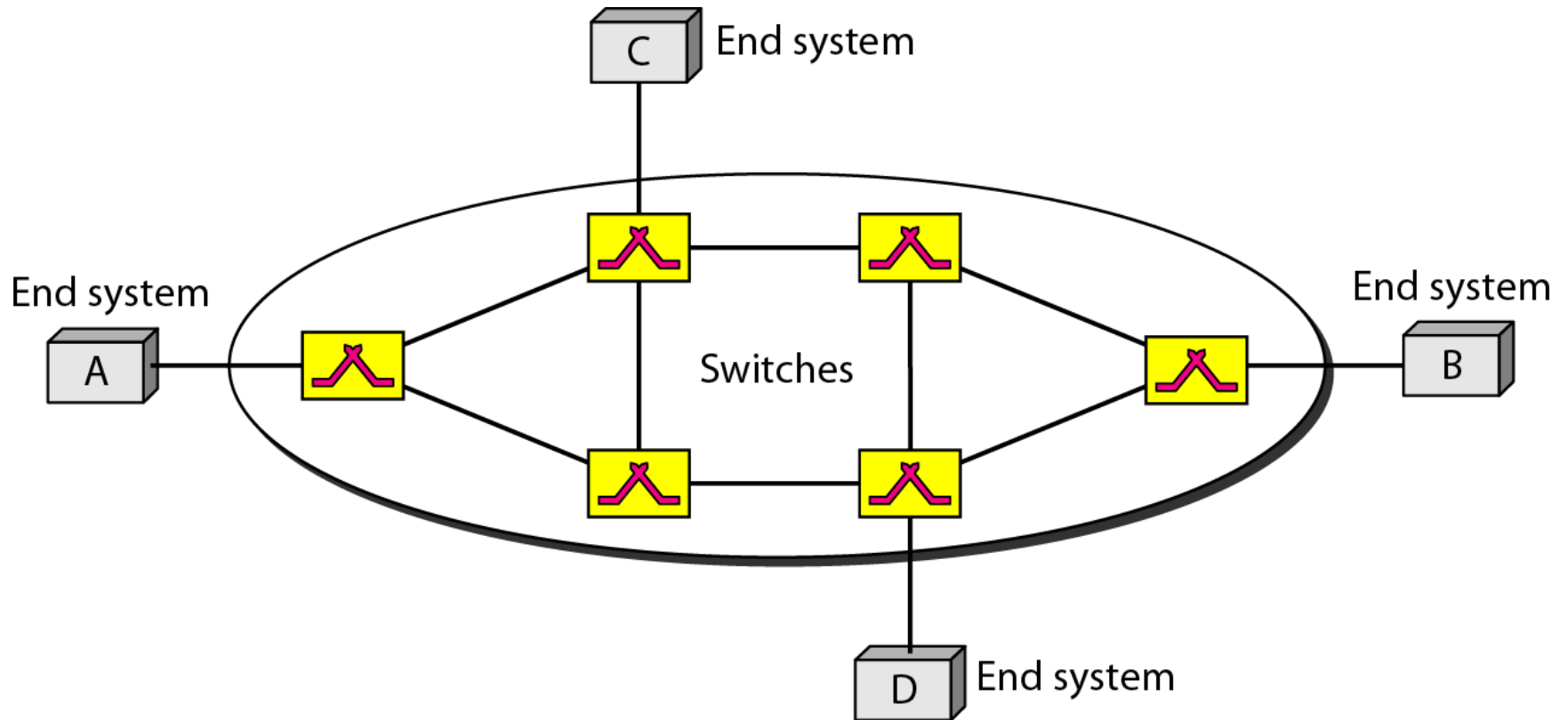
Note

*Switching in the Internet is done by using the datagram approach
to packet switching at
the network layer.*

8-3 VIRTUAL-CIRCUIT NETWORKS

A virtual-circuit network is a cross between a circuit-switched network and a datagram network. It has some characteristics of both.

Figure 8.10 *Virtual-circuit network*



A source or destination can be :

a computer, packet switch, bridge, or any other device that connects other networks.

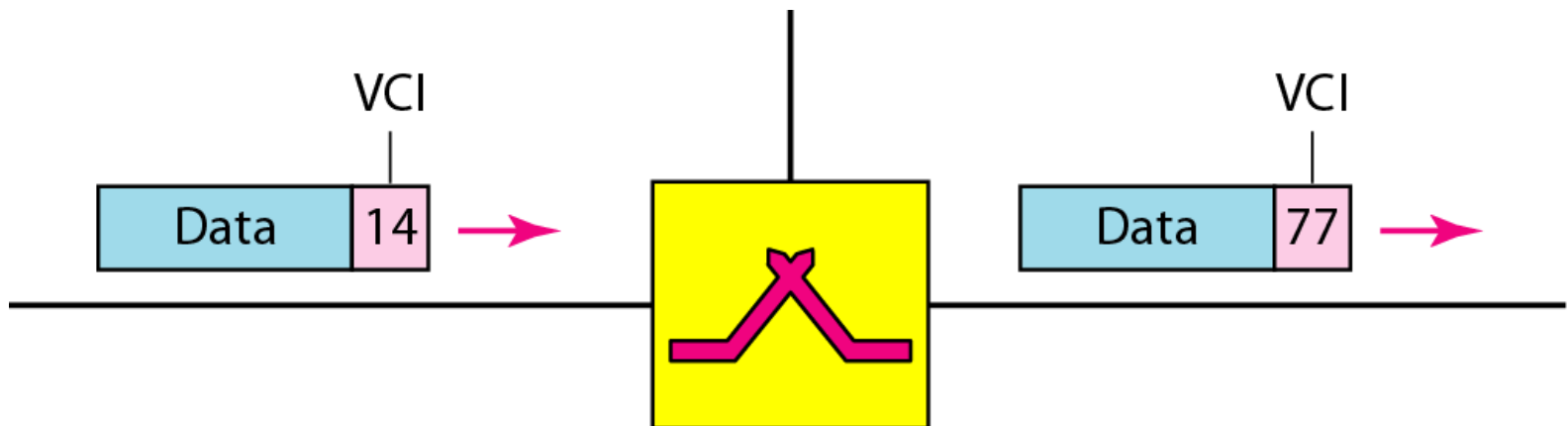
VIRTUAL-CIRCUIT NETWORKS- common characteristics

1. setup and teardown phases in addition to the data transfer phase(?).
2. Resources can be allocated during the setup phase(?), or on demand (?).
3. As in a datagram network, data are packetized and each packet carries an address in the header. However, the address in the header has **local jurisdiction** (it defines what should be the next switch and the channel on which the packet is being carried), **not end-to-end jurisdiction**.
4. As in a circuit-switched network, all packets follow the same path established during the connection.
5. A virtual-circuit network is normally implemented in the data link layer

Addressing

Two types of addressing are involved:

- Global (unique in the scope of the network)
- local (virtual-circuit identifier)
 - a small **number** that has only switch scope;
 - When a frame arrives at a switch, it has a VCI; when it leaves, it has a different VCI.



Three Phases

1. *Data Transfer Phase*
2. *Setup Phase*
3. *Teardown Phase*

(Pg 224, 225, 226 Forouzan)

Figure 8.12 *Switch and tables in a virtual-circuit network*

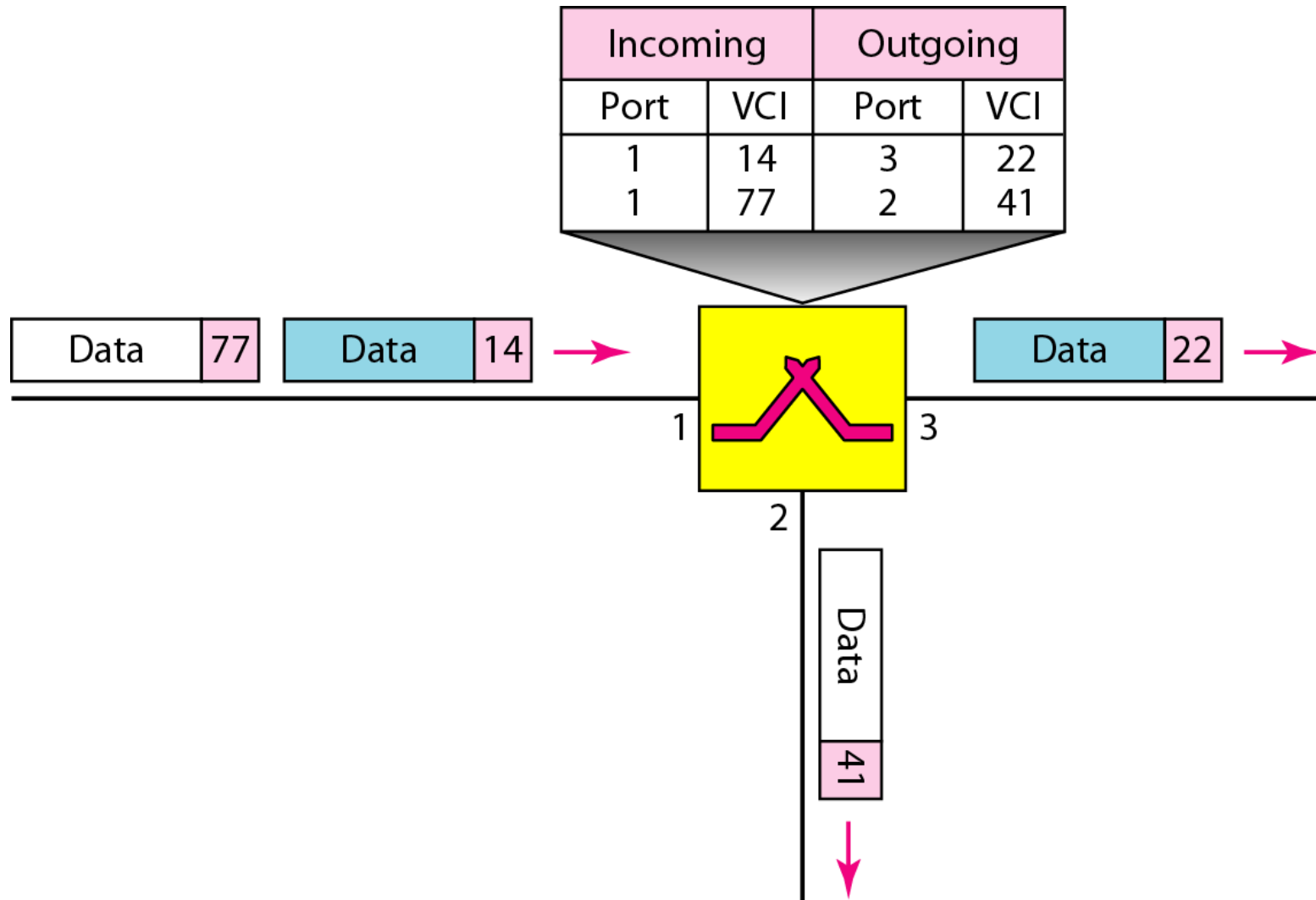


Figure 8.13 *Source-to-destination data transfer in a virtual-circuit network*

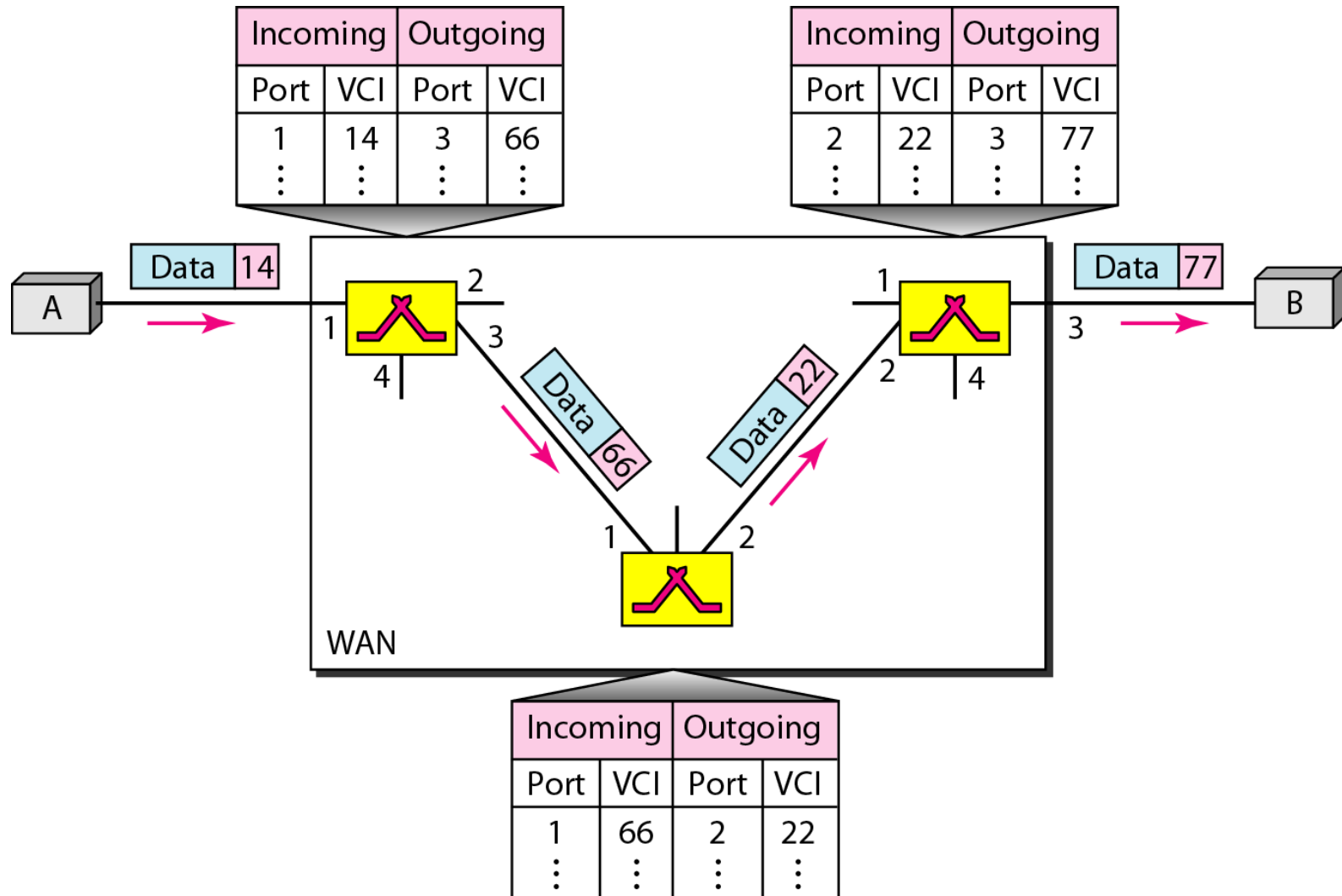


Figure 8.14 Setup request in a virtual-circuit network

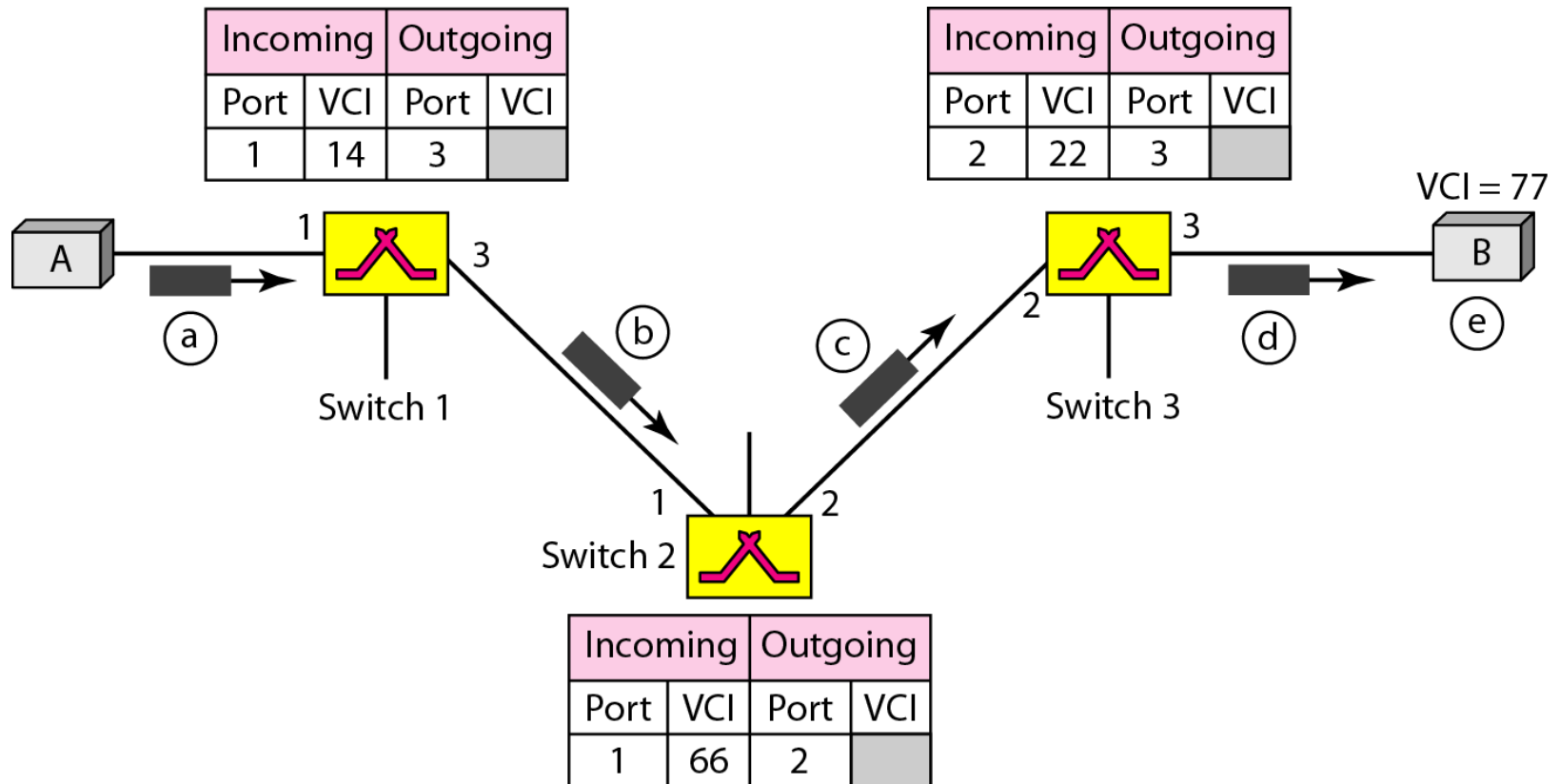
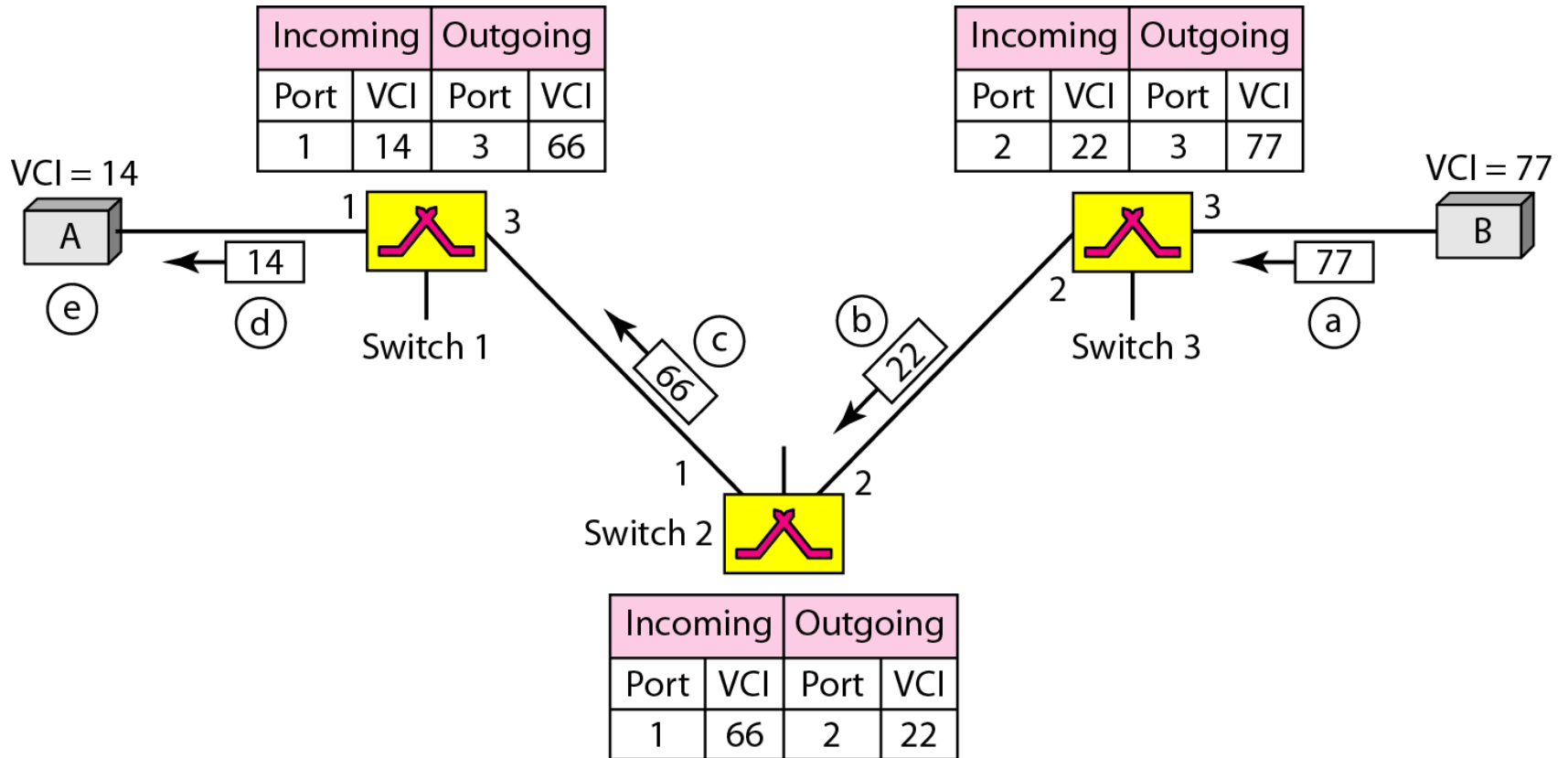


Figure 8.15 Setup acknowledgment in a virtual-circuit network



Note

In virtual-circuit switching, all packets belonging to the same source and destination travel the same path; but the packets may arrive at the destination with different delays if resource allocation is on demand.

Note

Switching at the data link layer in a switched WAN (Frame Relay and ATM networks.) is normally implemented by using virtual-circuit techniques.

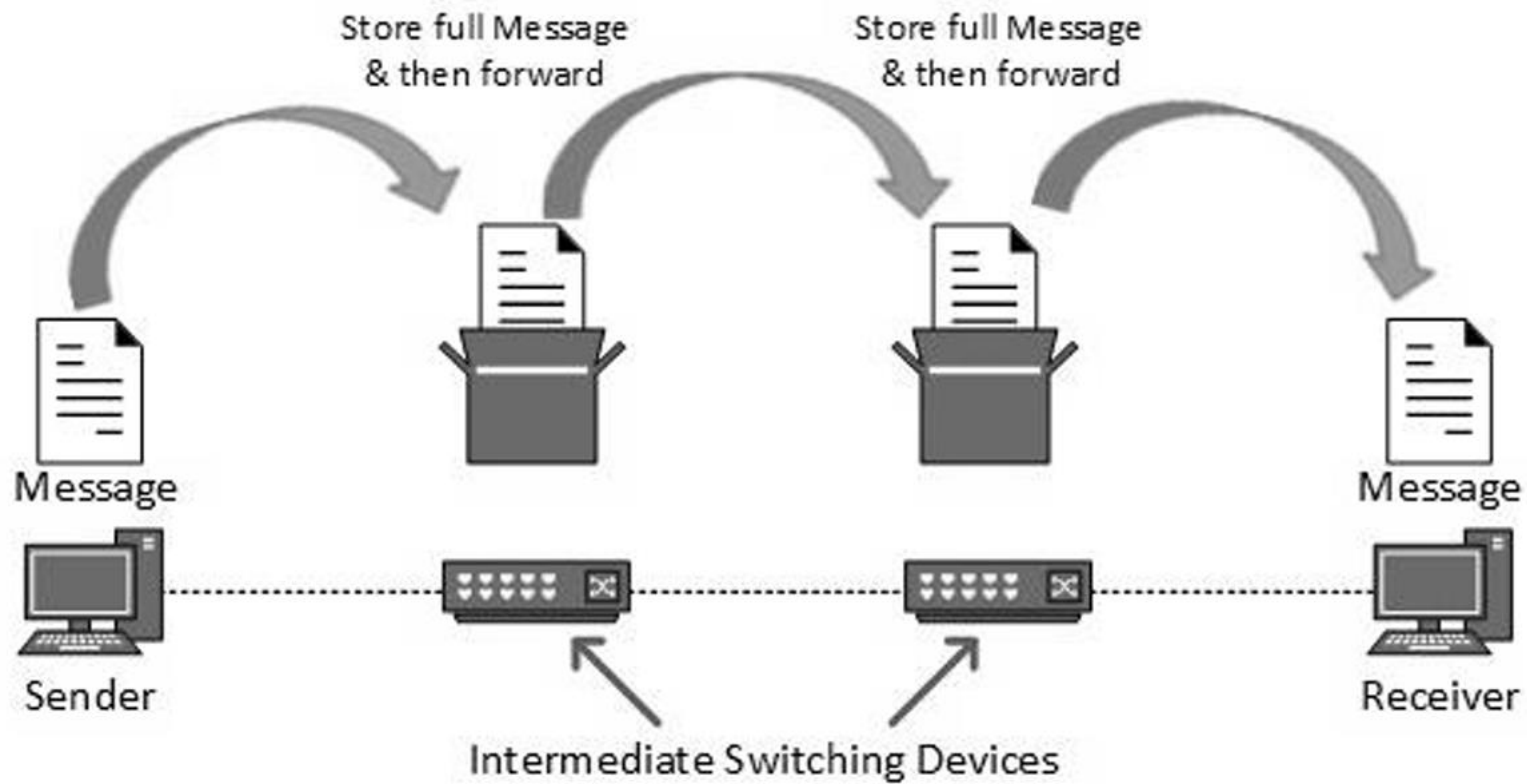
Comparison of communication switching techniques

Circuit Switching	Datagram Packet Switching	Virtual Circuit Packet Switching
Dedicated transmission path	No dedicated path	No dedicated path
Continuous transmission of data	Transmission of packets	Transmission of packets
Fast enough for interactive	Fast enough for interactive	Fast enough for interactive
Messages are not stored	Packets may be stored until delivered	Packets stored until delivered
The path is established for entire conversation	Route established for each packet	Route established for entire conversation
Call setup delay; negligible transmission delay	Packet transmission delay	Call setup delay; packet transmission delay
Busy signal if called party busy	Sender may be notified if packet not delivered	Sender notified of connection denial
Overload may block call setup; no delay for established calls	Overload increases packet delay	Overload may block call setup; increases packet delay
Electromechanical or computerized switching nodes	Small switching nodes	Small switching nodes
User responsible for message loss protection	Network may be responsible for individual packets	Network may be responsible for packet sequences
Usually no speed or code conversion	Speed and code conversion	Speed and code conversion
Fixed bandwidth	Dynamic use of bandwidth	Dynamic use of bandwidth
No overhead bits after call setup	Overhead bits in each packet	Overhead bits in each packet

Message Switching

- This technique was somewhere in middle of circuit switching and packet switching.
- In message switching, the whole message is treated as a data unit and is switching / transferred in its entirety.
- A switch working on message switching, first receives the whole message and buffers it until there are resources available to transfer it to the next hop.
- If the next hop is not having enough resource to accommodate large size message, the message is stored and switch waits.

Message Switching



Message Switching

- This technique was considered substitute to circuit switching.
- As in circuit switching the whole path is blocked for two entities only.
- Message switching is replaced by packet switching.

Message Switching-Drawbacks

- Message switching has the following drawbacks:
 - Every switch in transit path needs enough storage to accommodate entire message.
 - Because of store-and-forward technique and waits included until resources are available, message switching is very slow.
 - Message switching was not a solution for streaming media and real-time applications.

Routing

- A Router is a process of **selecting path** along which the data can be transferred from source to the destination. Routing is performed by a special device known as a **router**.
- A **Router** works at the **network layer** in the OSI model and internet layer in TCP/IP model
- The routing algorithms are used for routing the packets.
- The **routing algorithm** is nothing but a software responsible for deciding the optimal path through which packet can be transmitted.

WHAT IS ROUTING

- **Routing** is the task of selecting paths in a network along which to send **network traffic** or **information**
- Routing enables us to connect two or more networks using different network addresses i.e. Internetworking

WHAT IS ROUTER

- A computer or a device that specialize in sending packets over the data network
- Responsible for interconnecting networks
- Forwards packets to their destination by selecting the best path for a packet to travel

ROUTING TABLES

- The routing process usually directs forwarding on the basis of ***routing tables*** which maintain a record of the routes to various network destinations

ROUTING ACTIVITIES

- Routing involves the following two activities:
 - Path determination
 - Packet forwarding

PATH DETERMINATION

- The process of obtaining path in routing table is called **path determination**
- Path selection involves applying a routing metric to multiple routes, in order to select (or predict) the best route

PATH DETERMINATION

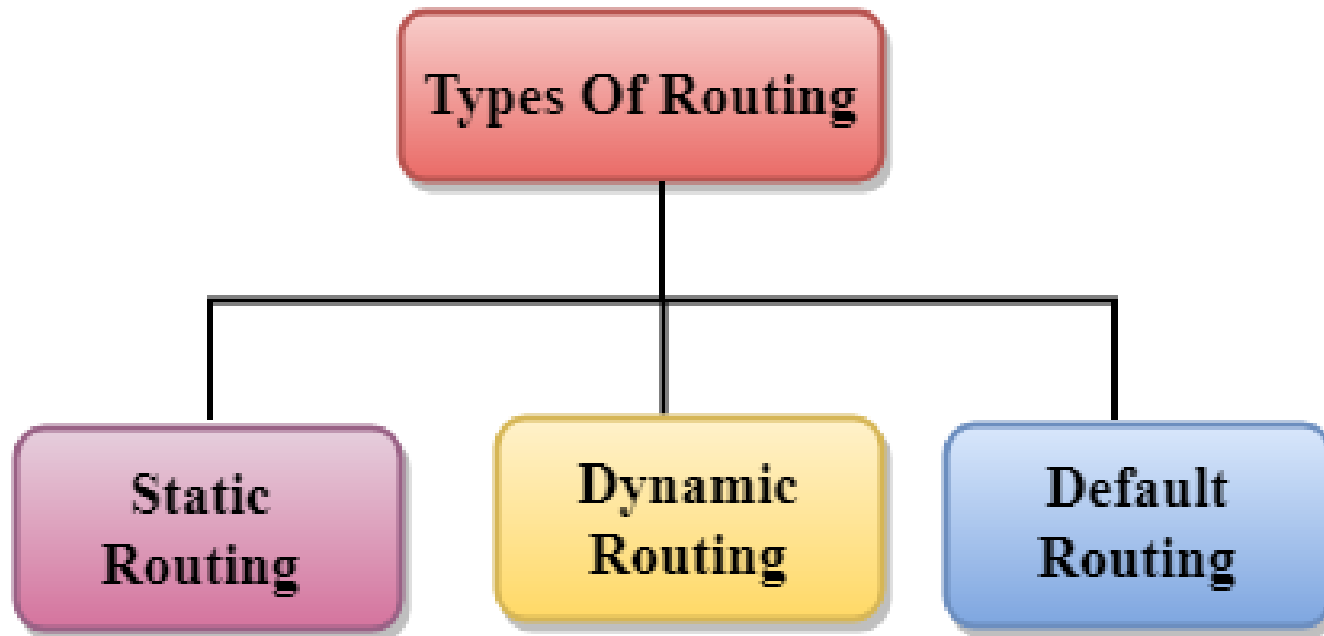
- There are three different methods to which router can learn path.
 - Automatic detection of directly connected network.
 - Static & Default routing
 - Dynamic routing

PACKET FORWARDING

- If a path is established using the information from the **routing table/s**, then the concerned router sends the packet to the next device on the selected path
- The router will perform packet forwarding only if route is available in the routing table

Types Of Routing

- Routing can be classified into three categories:-



1. Static Routing

- Static Routing is also known as Nonadaptive Routing.

It is a technique in which the administrator manually adds the routes in a routing table.

A Router can send the packets for the destination along the route defined by the administrator.

In this technique, routing decisions are not made based on the condition or topology of the networks

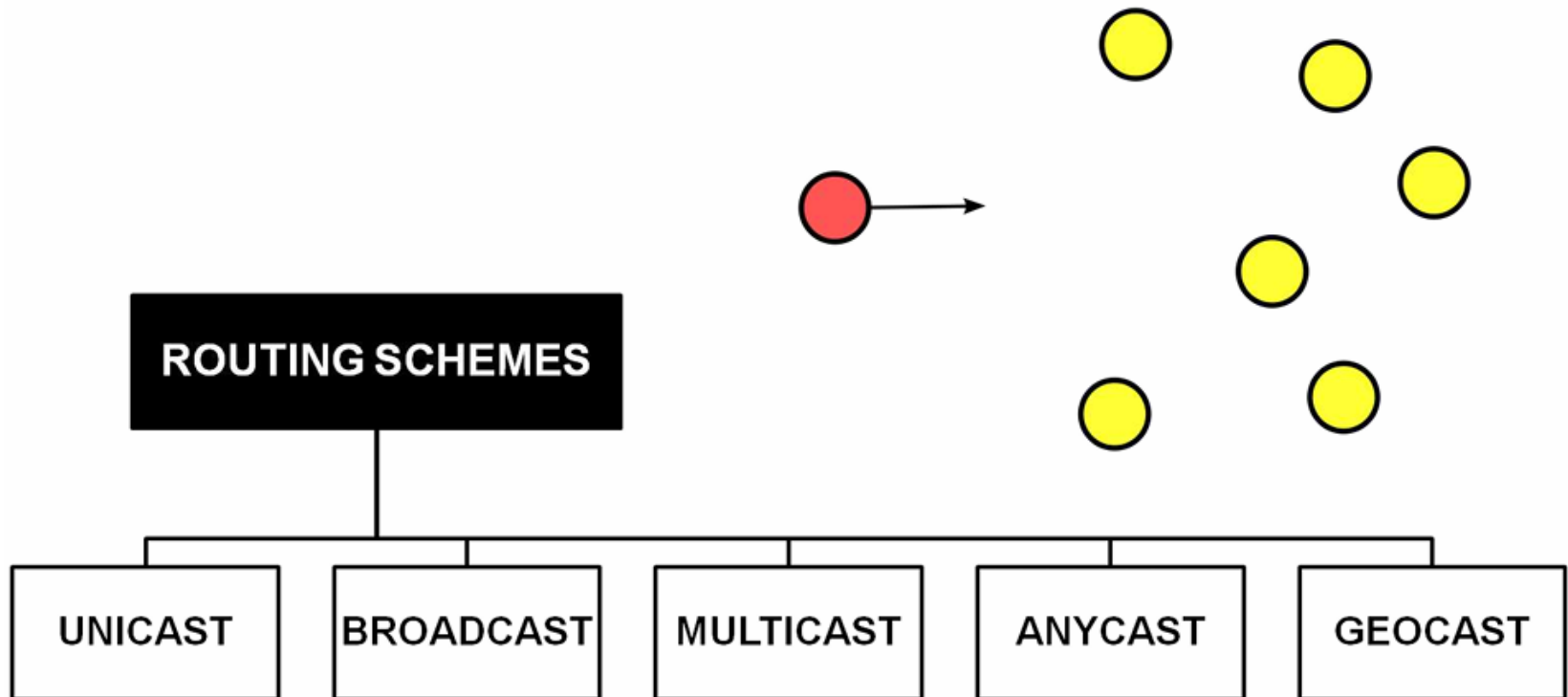
2. Dynamic Routing

- It is also known as Adaptive Routing.
- It is a technique in which a router adds a new route in the routing table for each packet in response to the changes in the condition or topology of the network.
- Dynamic protocols are used to discover the new routes to reach the destination.
- In Dynamic Routing, (Routing Information Protocol) RIP and (Open Shortest Path First) OSPF are the protocols used to discover the new routes.
- If any route goes down, then the automatic adjustment will be made to reach the destination.

3. Default Routing

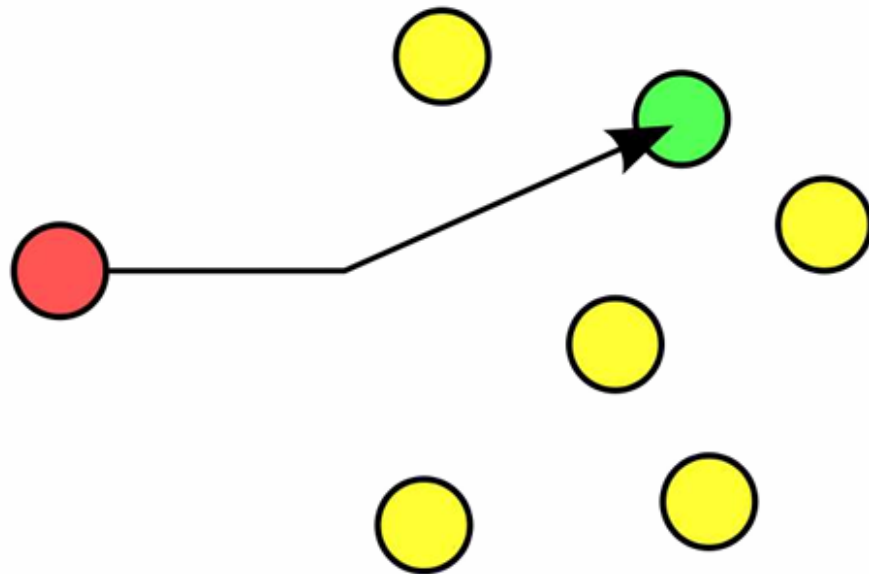
- Default Routing is a technique in which a router is configured to send all the packets to the same hop device, and it doesn't matter whether it belongs to a particular network or not. A Packet is transmitted to the device for which it is configured in default routing.
- Default Routing is used when networks deal with the single exit point.
- It is also useful when the bulk of transmission networks have to transmit the data to the same hp device.

ROUTING SCHEMES



ROUTING SCHEMES

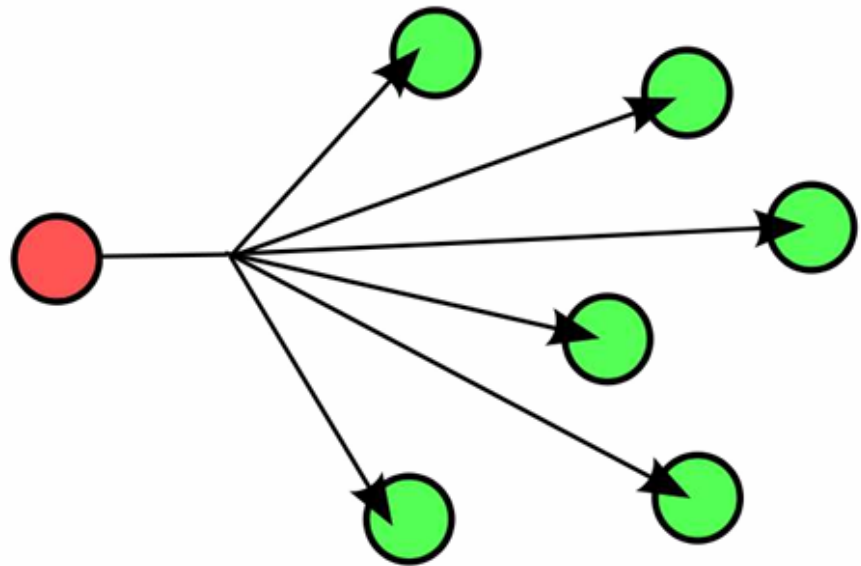
- **UNICAST** - delivers a message to a single specific node



Direct transmission from one sender to one specific recipient. It is the standard for point-to-point communication.

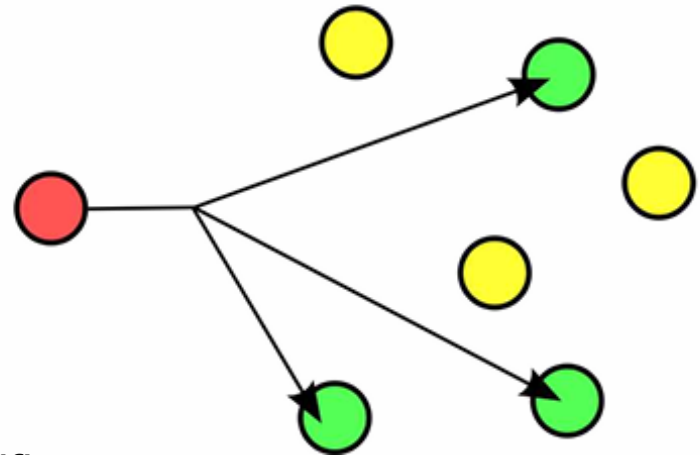
ROUTING SCHEMES

- **BROADCAST** - delivers a message to all nodes in the network



ROUTING SCHEMES

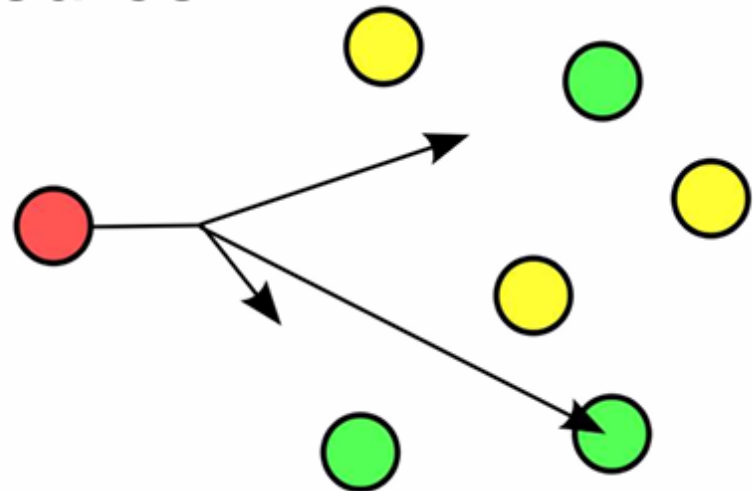
- **MULTICAST** - delivers a message to a group of nodes that have expressed interest in receiving the message



A single sender transmits data to a specific group of subscribers, such as in video conferencing

ROUTING SCHEMES

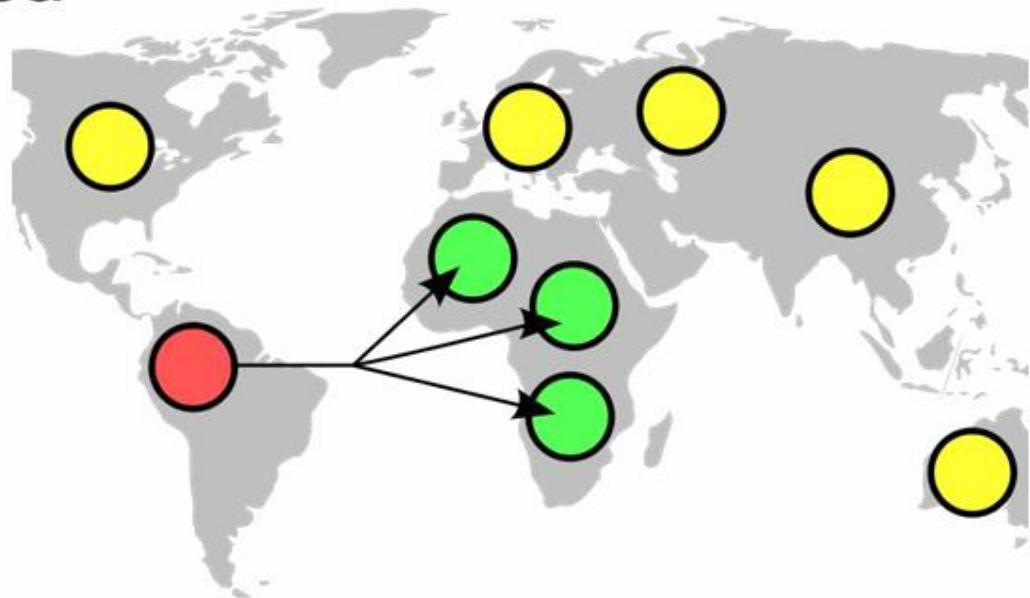
- **ANYCAST** - delivers a message to any one out of a group of nodes, typically the one nearest to the source



Data is routed to the closest or most efficient node among a group of servers sharing the same IP address,

ROUTING SCHEMES

- **GEOCAST** - delivers a message to a geographic area



A specialized routing method where data is delivered to a specific geographical area rather than a specific network address.

Glossary

T1

- Data transfer system that transfers digital signals at 1.544 megabits per second
- Large bandwidth -
 - Hence Most small to mid-sized colleges and business have T1 lines for their Internet connections
 - T1 line can cause dramatic decreases in data transfer speeds.

ISDN (Integrated Services Digital Network)

- data transfer technology that can transfer data significantly faster than a dial-up modem
- enables wide-bandwidth digital transmission over the public telephone network
- typical ISDN connection can support transfer rates of 64K or 128K of data per second
- newer DSL technology can support even faster transfer & is less costly to set

DSL (Digital Subscriber Line)

- medium for transferring data over regular phone lines and can be used to connect to the Internet
- much faster than a regular phone connection
- An **asymmetric DSL (ADSL)** allows download speeds of up to about 1.5 megabits (not megabytes) per second, and upload speeds of 128 kilobits per second
- Symmetric Digital Subscriber Line (SDSL), allows data transfer speeds of 384 Kilobits per second in both directions.

Bandwidth

- **refers to how much data you can send through a network or modem connection. It is usually measured in bits per second, or "bps."**
- think of bandwidth as a “**highway with cars travelling on it**”.

Throughput

- **Throughput refers to how much data can be transferred from one location to another in a given amount of time.**
- used to measure the performance of hard drives and RAM, as well as **Internet and network connections**.
- **Ex:**
 - A hard drive transfer rate of 100 Mbps has twice the throughput of a drive that can only transfer data at 50 Mbps.
 - 54 Mbps wireless connection has roughly 5 times as much throughput as a 11 Mbps connection

